07 — Conclusion and Future Enhancements

Item	Details
Course	DBMS Project — Semester 5
Team	Sneha Angelin Jisho (PES2UG24CS507); Sowmya Ramesh (PES2UG24CS512)
Stack	React + Vite; Node.js + Express; MySQL; MongoDB
Repository	https://github.com/snehajisho/Satellite_Data_Provenance_Earth_Observatory-DBMS_Project

14. Conclusion

The Satellite Data Provenance and Earth Observation Repository demonstrates a combined database and web-application approach to managing satellite observation information. MySQL stores structured relational entities and enforces relationships and domain constraints, while MongoDB stores nested observation metadata, spectral and quality information, processing metadata and image references.

A Node.js/Express backend exposes REST endpoints and connects to both database systems. Authentication uses bcrypt password hashing and JWT-based authorization with USER and ADMIN roles. The React/Vite frontend presents Home, Satellites, Observations, Analytics, Provenance and Quality views.

The database implementation demonstrates constraints, indexes, views, stored procedures, a function and triggers. SQL and MongoDB query scripts demonstrate relational joins, aggregation and document filtering. The project also includes SQL, MongoDB and API test suites. The project team reported successful execution of these suites.

The system serves as an academic DBMS demonstration. Sample observations, processing activities and station records should not be interpreted as independently validated operational satellite acquisition data.

14.1 Limitations

- The repository contains curated sample data rather than a continuously ingested archive of verified satellite acquisitions.
- MySQL and MongoDB are separate stores accessed through application code; no automatic replication or distributed cross-database transaction mechanism is implemented.

- The frontend repository does not include a dedicated automated component or end-to-end test framework; frontend checks are manual.
- Image previews depend on valid image files and paths.
- Production deployment concerns such as monitoring, backup/restore, audit logging and hardened access control require further work.

14.2 Future Enhancements

- Integrate trusted satellite data catalogue APIs and ingestion pipelines for verified acquisition metadata.
- Introduce controlled synchronization or event-driven workflows for changes spanning MySQL and MongoDB.
- Add automated frontend unit, component and end-to-end tests.
- Add spatial search, map-based exploration, temporal filters and richer geographic queries.
- Add pagination, server-side filtering and caching for larger datasets.
- Extend provenance with reproducible processing parameters, software versions, operator identity and audit history.
- Add monitoring, structured logging, backup/restore procedures and production deployment configuration.
- Strengthen production security with HTTPS, secret management, token revocation/session controls and rate limiting.

14.3 Closing statement

The project establishes an extensible foundation for an Earth-observation information repository. Its combination of relational integrity, flexible metadata storage, API-based access and provenance-oriented presentation can be developed into a more comprehensive satellite data management platform.